

Lineus longissimus Functional Annotation Report

Enhancement of NCBI Annotation Release GCF_910592395.1-RS_2024_04

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1 Organism Information

Field	Value
Species	<i>Lineus longissimus</i> (ribbon worm)
Taxonomy ID	88925
Common name	Bootlace worm
Phylum	Nemertea
BioSample	SAMEA7522833
BioProject	PRJEB45696

2 Assembly Summary

Field	Value
Assembly name	tnLinLong1.2
Submitter	Wellcome Sanger Institute
Assembly date	2022-02-13
Assembly type	Haploid (principal pseudohaplotype of diploid)
Assembly level	Chromosome
Genome representation	Full
RefSeq accession	GCF_910592395.1
GenBank accession	GCA_910592395.2
Alternate haplotype	GCA_910592375.2
RefSeq category	Reference Genome
Sequencing technology	PacBio, Illumina, Arima (Hi-C)
Genome coverage	79x
WGS project	CAJUZJ02

2.1 Assembly Statistics

Statistic	Value
Total length	391,165,532 bp (~391 Mb)
Ungapped length	391,151,032 bp
Total gap length	14,500 bp
GC content	41.5%
Chromosomes	19
Total scaffolds	30
Scaffold N50	20,961,375 bp (~21.0 Mb)
Scaffold L50	8
Scaffold N75	16,850,461 bp
Scaffold N90	16,292,257 bp
Contigs	107

Statistic	Value
Contig N50	9,910,080 bp (~9.9 Mb)
Contig L50	14
Spanned gaps	77
Unplaced scaffolds	10 (702,340 bp total)
Unlocalized scaffolds	1 (33,576 bp; on Chr 17)
Mitochondrial genome	15,284 bp (OU343011.1)

2.2 Chromosome Sizes

Chromosome	RefSeq Accession	Length (bp)	Gaps
1	NC_088308.1	29,445,314	0
2	NC_088309.1	29,188,318	1
3	NC_088310.1	29,083,594	6
4	NC_088311.1	25,598,605	4
5	NC_088312.1	23,943,388	4
6	NC_088313.1	21,663,950	5
7	NC_088314.1	21,159,187	6
8	NC_088315.1	20,961,375	6
9	NC_088316.1	20,549,149	0
10	NC_088317.1	19,406,143	2
11	NC_088318.1	19,191,461	2
12	NC_088319.1	17,976,742	3
13	NC_088320.1	16,850,461	7
14	NC_088321.1	16,759,716	6
15	NC_088322.1	16,292,257	16
16	NC_088323.1	16,534,156	1
17	NC_088324.1	16,304,662	3
18	NC_088325.1	15,888,539	5
19	NC_088326.1	13,632,599	0

2.3 Contamination Screening

FCS (Foreign Contamination Screen) v0.5.4 reported **no contamination** detected. The asserted division was `anml:worms`, with inferred primary divisions including molluscs and brachiopods (consistent with lophotrochozoan affinity).

2.4 Genomic Masking

22.93% of the assembly was masked using WindowMasker.

3 Structural Annotation

Pipeline: NCBI Eukaryotic Genome Annotation Pipeline v10.2 **Annotation release:** GCF_910592395.1-RS_2024_04 **Submission date:** 2024-05-01

3.1 Gene Counts

Feature	Count
Total genes & pseudogenes	21,203
Protein-coding genes	17,976
lncRNA genes	1,716
rRNA genes	477
tRNA genes	501
snRNA genes	319
snoRNA genes	79
Other ncRNA genes	6
Pseudogenes	129

3.2 Transcript and CDS Counts

Feature	Count
mRNAs	34,875
CDS (with protein)	34,875
lncRNAs	2,161
snRNAs	319
snoRNAs	79
scaRNAs	6
rRNAs	477
tRNAs	501
misc_RNAs	659

3.3 Transcript Statistics

Metric	Value
Mean transcripts per gene	1.88
Median transcripts per gene	1
Mean exons per transcript	11.71
Median exons per transcript	8
Total exons	183,038
Mean exon length	367 bp
Median exon length	151 bp
Single-exon coding transcripts	1,578

3.4 Annotation Quality

Metric	Value
Fully-supported mRNAs	31,733 / 34,875 (91.0%)
mRNAs with >5% ab initio	2,524 / 34,875 (7.2%)

4 Functional Annotation

4.1 BUSCO Assessment (metazoa_odb10)

Category	Percentage
Complete	98.2%
Complete & single-copy (S)	97.7%
Complete & duplicated (D)	0.5%
Fragmented (F)	0.4%
Missing (M)	1.4%

The very high BUSCO completeness (98.2%) indicates an excellent and near-complete gene space representation.

4.2 Protein Alignment to UniProtKB/Swiss-Prot

Coverage threshold	Genes	Percentage
>= 50% query coverage	12,024	66.9%
>= 95% query coverage	2,843	15.8%
100% query coverage	242	1.3%

4.3 Gene Naming and Characterization

Of 21,074 genes with descriptions in the GFF3 annotation:

Category	Genes	Percentage
Uncharacterized / hypothetical	7,344	34.9%
Named by homology (“-like” suffix)	11,647	55.3%
Confidently named (non-homology)	~2,083	~9.9%

The majority of gene names are inferred by sequence similarity to orthologs in other species, which is typical for a non-model lophotrochozoan lacking prior experimental gene characterization.

4.4 Gene Ontology (GO) Annotations

Overview: 56,362 GO annotations across 13,318 unique genes (74.1% of 17,976 protein-coding genes), using 5,487 unique GO terms.

All annotations are **IEA (Inferred from Electronic Annotation)** – none are experimentally validated.

Annotation sources:

Source	Annotations	Percentage
PANTHER (v18.0)	34,088	60.5%
InterPro (v5.65-97.0)	18,445	32.7%
Other	3,829	6.8%

Breakdown by GO ontology aspect:

Aspect	Annotations	Percentage
Molecular Function (F)	22,092	39.2%
Biological Process (P)	19,039	33.8%
Cellular Component (C)	15,231	27.0%

Among GO-annotated genes, 2,442 remain “uncharacterized” yet received GO terms through domain/family matches.

4.5 Gene Expression

Expression was quantified across 8 RNA-Seq datasets (see Section 6.1):

Category	Genes	Percentage
Expressed (>0 reads in at least one sample)	19,477	91.9%
No detected expression	1,726	8.1%
Total	21,203	100%

4.6 Functional Annotation Summary (NCBI baseline)

The NCBI baseline functional annotation is entirely computationally derived, relying on InterPro domain scans, PANTHER family classification, and Swiss-Prot protein alignments. While ~74% of protein-coding genes received at least one GO term, ~35% remain “uncharacterized” and over half are named only by homology.

5 Enhanced Functional Annotation (OSX 2026-03-31)

To improve upon the NCBI baseline, we ran a suite of additional annotation tools on the 34,875 predicted proteins. All analyses were performed locally (2026-03-31).

5.1 eggNOG-mapper

Tool: eggNOG-mapper v2 with eggNOG v5.0 database, tax_scope Metazoa

Metric	Count	% of 34,875 proteins
Proteins annotated	27,965	80.2%
With GO terms	22,273	63.9%
With KEGG KO	19,350	55.5%
With KEGG pathway	11,743	33.7%
With KEGG module	2,763	7.9%
With COG category	26,810	76.9%
With functional description	26,810	76.9%

GO term improvement: eggNOG-mapper assigned 21,450 unique GO terms (vs. 5,487 from NCBI baseline) to 22,273 proteins – a ~4x increase in GO vocabulary breadth.

COG functional categories:

COG	Description	Proteins
S	Function unknown	6,719
T	Signal transduction mechanisms	6,141
O	Post-translational modification, protein turnover	1,951
K	Transcription	1,889
U	Intracellular trafficking, secretion, transport	1,411
P	Inorganic ion transport and metabolism	1,377
Z	Cytoskeleton	1,256
G	Carbohydrate transport and metabolism	1,041
A	RNA processing and modification	874
L	Replication, recombination and repair	865
E	Amino acid transport and metabolism	858
I	Lipid transport and metabolism	771
J	Translation, ribosomal structure and biogenesis	676
D	Cell cycle control, cell division	502
C	Energy production and conversion	501
Q	Secondary metabolites biosynthesis	388
B	Chromatin structure and dynamics	365
V	Defense mechanisms	312
F	Nucleotide transport and metabolism	304
W	Extracellular structures	301
M	Cell wall/membrane/envelope biogenesis	290
H	Coenzyme transport and metabolism	173
N	Cell motility	85

Signal transduction (T) is the largest informative category (6,141 proteins).

5.2 KofamScan

Tool: KofamScan with KEGG HMM profiles

Metric	Count
Significant KO hits	18,234
Unique proteins with KO	17,552
Unique KOs assigned	6,459

5.3 Combined KEGG Coverage (eggNOG + KofamScan)

Metric	Count
Proteins with KEGG KO	23,523 (67.5% of 34,875)
Unique KOs	13,504
Unique KEGG pathways (eggNOG)	764
Unique KEGG modules (eggNOG)	262

This represents a complete KEGG pathway annotation from zero (no KEGG data existed in the NCBI baseline).

5.4 SignalP 6.0

Tool: SignalP 6.0h, organism: eukarya

Category	Count	% of proteins
Signal peptide detected	4,221	12.1%
No signal peptide	30,654	87.9%

5.5 dbCAN – CAZyme Annotation

Tool: dbCAN v4, methods: DIAMOND + HMMER + dbCANsub

Metric	Count
Total CAZyme hits	2,048
High-confidence (≥ 2 tools)	360

Top CAZyme families (≥ 2 tools supporting):

Family	Count	Function
GT27	29	Polypeptide N-acetylgalactosaminyltransferases
GH38	20	Alpha-mannosidases
GH31_1	17	Alpha-glucosidases
AA1	17	Multicopper oxidases (laccases)
GT14	14	Beta-1,3-galactosyltransferases

Family	Count	Function
GT10	12	Fucosyltransferases
AA3_2	12	Glucose/aryl-alcohol oxidases
GH65	11	Trehalases/maltose phosphorylases
GT98	10	Glycosyltransferases
GT13	10	Alpha-1,4-N-acetylglucosaminyltransferases

5.6 OrthoFinder – Comparative Genomics

Tool: OrthoFinder 3.1.3, DIAMOND, 10 species

Species panel:

Species	Group	Proteins
<i>Lineus longissimus</i>	Nemertea	34,875
<i>Lingula anatina</i>	Brachiopoda	41,392
<i>Crassostrea gigas</i>	Bivalvia	51,045
<i>Pecten maximus</i>	Bivalvia	39,918
<i>Lottia gigantea</i>	Gastropoda	23,822
<i>Octopus sinensis</i>	Cephalopoda	36,112
<i>Helobdella robusta</i>	Annelida	23,426
<i>Drosophila melanogaster</i>	Arthropoda	30,802
<i>Strongylocentrotus purpuratus</i>	Echinodermata	38,439
<i>Homo sapiens</i>	Chordata	136,807

Results for *L. longissimus*:

Metric	Count
Proteins in orthogroups	33,541 (96.2%)
Unassigned (species-unique singletons)	1,334 (3.8%)
Species-specific orthogroups	930 (4,573 proteins)
Orthogroups with all 10 species	4,169
Single-copy orthogroups (all species)	45
Total orthogroups across all species	27,503

Inferred species tree (Newick):

```
((Homo_sapiens,Strongylocentrotus_purpuratus),
 (Drosophila_melanogaster,
 (Helobdella_robusta,
 ((Lineus_longissimus,Lingula_anatina),
 (Octopus_sinensis,
 (Lottia_gigantea,
 (Crassostrea_gigas,Pecten_maximus)))))));
```

The species tree correctly recovers *L. longissimus* (Nemertea) as sister to *L. anatina* (Brachiopoda) within Lophotrochozoa, consistent with the Kryptrochozoa hypothesis. The 96.2% assignment rate to orthogroups confirms that the vast majority of *L. longissimus* genes have identifiable homologs across Metazoa.

5.7 Annotation Improvement Summary

Annotation type	NCBI baseline	Enhanced	Improvement
Proteins with GO terms	13,318 genes	22,273 proteins	+67% genes
Unique GO terms	5,487	21,450	+291%
KEGG KO assignments	0	23,523 proteins	new
KEGG pathways	0	764	new
KEGG modules	0	262	new
COG categories	0	26,810 proteins	new
Signal peptides	0	4,221 proteins	new
CAZymes (high-conf.)	0	360 proteins	new
Orthogroup assignment	0	33,541 (96.2%)	new

6 Structural Annotation Evidence Sources

6.1 RNA-Seq Data Used

8 RNA-Seq runs were used to support the annotation:

SRA Run	Sample	Reads	Alignment Rate	Spliced Rate	Platform
ERR9434983	Mid body	44,423,822	84%	30%	Illumina
SRR1324981	Whole body	7,442,212	85%	21%	Illumina
SRR1324982	Whole body	6,030,623	86%	20%	Illumina
SRR1324983	Whole body	6,033,086	85%	21%	Illumina
SRR1324984	Whole body	26,127,395	88%	31%	Illumina
SRR1324985	Whole body	28,304,265	88%	25%	Illumina
SRR1324986	Whole body	21,639,029	92%	29%	Illumina
SRR2682192	Embryo/juvenile mix	435,278,178	53%	17%	Illumina

Total reads used: ~575.3 million

6.2 Cross-species Protein Alignments

Proteins from the following species were aligned to support gene prediction: - *Homo sapiens* - *Strongylocentrotus purpuratus* (sea urchin) - *Drosophila melanogaster* (fruit fly)

7 Key Highlights

1. **Chromosome-level assembly** with 19 chromosomes, scaffold N50 of ~21 Mb, and only 14.5 kb total gaps – an exceptionally contiguous assembly.
 2. **17,976 protein-coding genes** with 34,875 mRNAs (mean 1.88 transcripts/gene), indicating moderate alternative splicing.
 3. **98.2% BUSCO completeness** (metazoa_odb10) confirms near-complete gene space capture.
 4. **91% of mRNAs are fully supported** by evidence (RNA-Seq and/or protein alignments), with only 7.2% containing significant ab initio prediction.
 5. **No contamination** detected by FCS screening.
 6. The annotation was produced by **NCBI Eukaryotic Genome Annotation Pipeline v10.2** (April 2024), representing the most current annotation available for this species.
 7. **Enhanced functional annotation** added KEGG pathway mapping (764 pathways, 23,523 proteins with KO), increased GO vocabulary 4x (21,450 unique terms), identified 4,221 signal peptides, 360 CAZymes, and placed 96.2% of proteins into orthogroups across 10 metazoan species.
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8 Data Provenance

- **Assembly FTP:** https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/910/592/395/GCF_910592395.1_tnLinLong1.2/
- **Annotation report:** https://www.ncbi.nlm.nih.gov/refseq/annotation_euk/Lineus_longissimus/GCF_910592395.1_RS_2024_04/
- **Annotation FTP:** https://ftp.ncbi.nlm.nih.gov/genomes/all/annotation_releases/88925/GCF_910592395.1_RS_2024_04/